GROUNDED DI OS

CONTROL AUDIT & HARNESS HARDENING

A companion to Two deterministic-control demos
How a local counterexample repaired DI² v0.3.1, and what the Microsoft mirror actually tested

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This second book records a bounded feedback loop: local counterexamples were used to repair Grounded DI's own DI² harness, and the resulting v0.3.1 candidate was then test-driven against an unchanged, pinned slice of Microsoft's Agent Governance Toolkit. The work is authorized local software testing of public code. It does not describe a live Microsoft service, a production incident, or a product ranking. Microsoft retains its own source attribution; Grounded DI / MSW authorship and DIP106 lineage attach to Grounded DI's materials, not to Microsoft code.

1. The sequence matters

Step	Executed record	Consequence
Earlier Microsoft demonstrations	The first book preserves screenshots reporting v3's 216 upstream passes, 11 skips, 16 targeted passing controls and 8 exceptions across five groups; later v4.3 captures report 14 failures across five groups.	Historical screenshot records, not executable inputs to this mirror. Keep their counts separate.
DI² v0.3 hardening review	A fresh local comparison found an expiry-check placement regression. The unchanged v0.3 replay could pass while targeted late-entry assertions failed.	A minimal v0.3.1 runtime repair moved the final finite-clock check after validation and consumption in memory and SQLite paths. No new architecture, schema or tier.
v0.3.1 verification	Two rounds each passed 105 core and 135 unit registrations. The 32-case expiry comparison yielded 13 intended failures on v0.3 and 32 passes on v0.3.1. The six deliberately defective fixtures remain property failures.	Local candidate classification: synthetic hardening candidate with optional persistent local gate. No production claim.
New Microsoft mirror	Commit `bb9fd0df2aed2969a403b6d083c34185cad0eb04` and 11 primary modules (3,184 physical lines). Eight upstream files: 216 passes, 11 OPA skips. Added local registry: 25 Microsoft passes, 13 failed safety assertions across five groups; all 13 reproduced in a fresh process.	A separate adapter and registry exercised the unchanged harness runner. No material v0.3.1 defect requiring repair was demonstrated.
DI² native comparison	MEMORY_ONLY: 26 passes. SQLITE_LOCAL: 27 passes. Twenty-two fixture intents were common, with declared representation and configuration differences.	Positive evidence for tested local properties only; unsupported paths are not counted as passes.

The v0.3.1 repair preceded the final Microsoft mirror. This book does **not** claim that the 13 mirror failures caused that repair. It shows two legitimate uses of the same audit discipline: hardening one's own candidate against a reproduced regression, then testing its evidence infrastructure and the target on separately labeled native paths.

2. What repeated in the Microsoft path

These 13 cases are **five issue groups, not 13 independent vulnerabilities**:

01 **Incomplete effective argument binding — four cases.** The wrapper's context projection records the first positional action and keyword arguments but does not bind positional tails, variadic arguments or omitted callable defaults. Different actual callable-entry values can share an action digest. Source: `govern.py:892–963` at the pinned commit.

02 **Approved snapshot versus callable entry — three cases.** A permitted approval or advisory callback can change original nested arguments between review and invocation. In the registered sync and async cases, review saw `n=1`; the callable entered with `n=999`. Source: `govern.py:414–603, 794–840, 892–963`.

03 **Expiry sampled before remaining work — three cases.** Injected clocks and disclosed scheduling/store instrumentation placed callable entry at offset 20 after a deadline of 10. Exact-deadline submission correctly rejected in a control. This is a final-sample placement finding, not proof of strict remote or wall-clock timing. Source: `approval_protocol/coordinator.py:311–365`; `govern.py:433–603`.

04 **Ordered stages — one case.** Required stage 1 was accepted before stage 0; after both identities participated the chain resolved allow and the local callable entered once. This is an order violation, not evidence that one identity satisfied both stages. Source: `approval_protocol/coordinator.py:191–285`; ADR-0030 section 4.

05 **Conflicting submissions — two cases.** Conflicting reuse of an event ID returned the original allow without a content comparison; a fresh ID appended a conflicting same-stage deny. The latter remained fail-closed. Neither case showed unauthorized invocation. The same-ID payload rule is a stronger audit expectation; the ADR expressly addresses same-stage conflict. Source: `approval_protocol/coordinator.py:211–285`.

Contrary controls passed: ordinary authorized calls, keyword-only and nested JSON distinction, missing-stage and unpermitted-identity denial, policy/action/chain mismatch rejection, exact duplicate delivery, advisory nonauthority, malformed-response denial, competing consumption, and configured advisory behavior. This is why the issue groups are stated narrowly.

3. How the harness was boosted

The v0.3.1 change repaired one reproduced local regression: expensive validation could consume the authorization interval after an earlier clock sample. In the original controlled memory comparison, unchanged v0.3 entered at 23.674 ms and 102.419 ms with a 3 ms interval; v0.3.1 denied without invocation on the same inputs. A 32-case targeted comparison, real-clock stress, crash/receipt checks, mutation checks, and byte-repeatability checks supported the repair within the stated local contract. `runtime.py` changed by 15 added and two removed lines including comments; `durable.py` did not change.

The Microsoft mirror then tested the **new harness as audit infrastructure**. Its shipped integrity and full replay passed. A separately written adapter ran through the unchanged candidate `unittest_runner.py` and `reporting.classify`, while Microsoft and DI² native results stayed separate. All 91 registered cases reconciled; the 13 Microsoft safety failures were assertion failures rather than crashes, repeated in a fresh process and checked by a second witness algorithm against callable-entry and event observations. Five observation-fixture mutations exercised the new oracle; they were not mutations of Microsoft's production code.

One early audit assertion about omitted versus explicit defaults was too strong and was withdrawn; the corrected case observed mutable default value at actual entry. A transport-added newline in 39 dependency copies was corrected to exact pinned Git blobs, and the final upstream and native suites were rerun. These are audit-method corrections, not harness repairs or Microsoft edits. The source, prior plan and raw results remain retained.

The practical maintenance rule is to **hold v0.3.1**. A new counterexample warrants a targeted regression case and the smallest justified correction, not an automatic architecture upgrade. A local pass earns a bounded claim; it does not silently expand supported surfaces.

4. Determinism and the actual claim

A stable digest can omit an argument. That is **incomplete binding**, not computational nondeterminism. Deterministic policy evaluation, canonical byte repeatability, workflow correctness, and enforcement through callable entry are distinct properties. The mirror's DL labels apply the six definitions in the v0.3.1 package only to registered boundaries: Microsoft local DL1 PARTIAL, DL2 FAIL, DL3 PARTIAL, DL4 UNTESTED, DL5 FAIL, DL6 FAIL; DI² local DL1 PARTIAL, DL2 PASS, DL3 PASS, DL4 UNTESTED, DL5 PASS, DL6 PASS. Those cells are not whole-product or cross-model certifications. The first book's historical screenshot DL assessment is a separate record with a different evidence base.

DI² general async execution, real identity authentication, complete closure/global/environment binding, external exactly-once effects, strict hard-real-time deadlines and production readiness remain unsupported or untested here. Microsoft's durable-store recovery, real remote identity, live endpoints and external effects were not tested. Eleven upstream OPA tests skipped because the CLI was unavailable. The host's pydantic 2.14.0 was below the full dev-extra's declared 2.14.1 minimum. No Microsoft-wide inference, incident frequency, patent infringement, commercial superiority or real-world effect is established.

5. Evidence ledger and replay

Record	Exact identity / role
First book	Screenshot-based historical companion; unchanged. The public text record is available separately.
DI² harness archive	Original local v0.3.1 candidate, retained within the mirror evidence. Full harness materials are not included in this edition.
DI² repair report	Historical repair report, raw evidence and remaining limits are retained in the research record.
Microsoft mirror report	Historical mirror report retains the case matrix, source references, observations and corrections.

Record	Exact identity / role
Complete mirror evidence	The historical mirror archive contains the evidence required for a full local replay. It is not included in this public edition.
Microsoft source	`https://github.com/microsoft/agent-governance-toolkit` at commit `bb9fd0df2aed2969a403b6d083c34185cad0eb04`; retained 95 source files verified against Git blob IDs. The MIT/copyright notices remain with those files.

The full replay requires the separately retained mirror evidence and controlled local inputs. The first book's nine image captures and the historical v4.3 executable inputs are not included here; the v4.3 package was not replayed in the mirror. This book is an editorial synthesis of the historical record, not a replacement for raw evidence.

Disposition: Retain v0.3.1 as the local hardening candidate and maintain the issue-group report and regression matrix. Update the record for a changed pinned commit, new root cause, tested repair or materially different boundary.